



COMSATS University Islamabad Vehari

Department of Computer Science

Assignment 3 – Spring 2026

Subject: Programming Fundamentals (CSC-103)

Instructor: Dr. Muhammad Mudassar

Class BSCS Batch SP26

Marks=10

CLO-1 “Demonstrate the fundamental concepts of programming.”

CLO-2 “Employ programming constructs using a programming language.”

Instructions:

1. Create an integrated PDF of programs, outputs, and answers.
2. Use CamScanner or similar software for PDF creation.
3. It is mandatory to upload the PDF to the CUOnline portal.
4. Deadline: 08 May 2026 till 10:00 AM.

1	The date June 10, 1960, is special because when it is written in the following format, the month times the day equals the year: 6/10/60 Design a program using method that asks the user to enter a month (in numeric form), a day, and a two-digit year (use these values as parameters to the method). The program should then determine whether the month times the day equals the year. If so, it should display a message saying the date is magic. Otherwise, it should display a message saying the date is not magic.
2	Factorial: In mathematics, the factorial of an integer n, denoted by $n!$ is the following product: $n!=1\times2\times...\times n$ For the given integer n calculate the value $n!$ use loop. a) Create a simple method that receives the value of n calculates the factorial and return the calculated factorial. Print this factorial in the main function as the result. b) Use Recursion to calculate the factorial.
3	Write a program that asks user to enter an ASCII code (an integer between 0 and 127) and displays: a) its character. b) Write a program that receives a character and displays its Unicode.
4	Write a program that prompts the user to enter an integer between 0 and 15 and displays its corresponding hex number.
5	Write a program that displays a random uppercase letter Use the <code>Math.random()</code> method (use integer value range of the ASCII code)

6	Write a method that computes the sum of the digits in an integer. Use the following method header: <pre>public static int sumDigits(long n)</pre> For example, sumDigits(234) returns 9 (2 + 3 + 4).
7	Write a method with the following header to display an integer in reverse order <pre>public static void reverse (int number)</pre> For example, reverse(3456) displays 6543. Write a test program that prompts the user to enter an integer and displays its reversal. Also check if the number is palindrome number or not.
8	Write a method with the following header to display three numbers in increasing order: <pre>public static void displaySortedNumbers (double num1, double num2, double num3)</pre> Write a test program that prompts the user to enter three numbers and invokes the method to display them in increasing order.
9	Write a method that counts the number of letters in a string using the following header: <pre>public static int countLetters(Strings)</pre> Write a test program that prompts the user to enter a string and displays the number of letters in the string.
10	Write a function capitalize(lower_case_word) that takes the lower case word and returns the word with the first letter capitalized. Eg., <pre>System.out.println(capitalize("word"))</pre> should print the word Word. Then, given a line of lowercase ASCII words (text separated by a single space), print it with the first letter of each word capitalized using the your own function capitalize().
11	Write a Java method to count all vowels in a string.

12	Given a positive real number a and a non-negative integer n. Calculate a^n without using loops, <code>**</code> operator or the built in function <code>math.pow()</code>. Instead, use recursion and the relation $a^n = a \cdot a^{n-1}$. Print the result. Form the method power(a, n)
13	The area of a pentagon can be computed using the following formula: $\text{Area} = (5 * s^2) / (4 * \tan(\pi/5))$ Write a method that returns the area of a pentagon using the following header: public static double area(double side) Write a main method that prompts the user to enter the side of a pentagon and displays its area.
14	Implement a Java code to perform method overloading for two methods named as reverse. First method reverses the provided number and second method reverses the letters in the provided string in the argument. <pre>public static void reverse (int num) { } public static void reverse (String subject) { }</pre> Write a main method where ask the user to enter year of his birth and store this in a number, also ask to enter a subject name store this in a string variable and call the reverse method accordingly. In the methods reverse the number (year) and subject and finally print these results.
15	Write a Java program that demonstrates method overloading by calculating the area of different geometric shapes. Implement the following overloaded methods: calculateArea(double radius) → Returns the area of a circle (Use formula: $\text{Math.PI} * \text{radius} * \text{radius}$). calculateArea(double length, double width) → Returns the area of a rectangle (Use formula: $\text{length} * \text{width}$). calculateArea(double base, double height, String shape) → Returns the area of a triangle (Use formula: $0.5 * \text{base} * \text{height}$). Requirements: - Use method overloading (same method name, different parameters).

	• The main() method should call each version of calculateArea() and display the results. • Use case statement in the main to decide about the area, and then ask the user to enter the required values and then call the area function accordingly.
16	Write a recursive method that returns the largest integer in an array. Write a test program that prompts the user to enter a list of eight integers and displays the largest element.
17	Write a recursive method to compute the following series $m(i) = \frac{1}{3} + \frac{2}{5} + \frac{3}{7} + \frac{4}{9} + \frac{5}{11} + \frac{6}{13} + \dots + \frac{i}{2i+1}$ Write a test program that displays m(i) for i = 1, 2, . . . , 10
18	Given a positive integer 10, the task is to find the 10th Fibonacci number using recursion. The Fibonacci sequence is a sequence where the next term is the sum of the previous two terms. The first two terms of the Fibonacci sequence are 0 followed by 1. The Fibonacci sequence: 0, 1, 1, 2, 3, 5, 8, 13, 21 Example: Input: n = 2 Output: 1 Explanation: 1 is the 2nd number of Fibonacci series. Input: n = 6 Output: 8 Explanation: 8 is the 6th number of Fibonacci series.
19	Write a recursive method that displays an int value reversely on the console using the following header: <pre>public static void reverseDisplay(int value)</pre> For example, reverseDisplay(12345) displays 54321. Write a test program that prompts the user to enter an integer and displays its reversal.
20	(Conversions between feet and meters) Write a class that contains the following two methods: <pre>/** Convert from feet to meters */ public static double footToMeter(double foot) /** Convert from meters to feet */ public static double meterToFoot(double meter)</pre>

	The formula for the conversion is:
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meter = 0.305 * foot

foot = 3.279 * meter

in the main function ask the user to enter the length and whether he entered foot or meters, then convert the length to other standard and finally print the results.